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Key Points:
- Primary magnetizations are preserved in the lower Beaufort Group in the Karoo Basin
- The magnetostratigraphy is compatible with a late Permian age for the Beaufort Group
- A re-examination of existing data from the underlying Ecca Group indicates a mid-Permian age

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Magnetostratigraphic constraints on the age of the lower Beaufort Group, western Karoo basin, South Africa, and a critical analysis of existing U-Pb geochronological data

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Abstract We studied three partially overlapping sections with a composite thickness of ~600 m in the upper Permian fluvial siltstones and fine-grained sandstones of the Abrahamskraal Formation, the basal unit of the Beaufort Group, in the Karoo Basin of Western Cape Province, South Africa. Paleomagnetic analysis reveals three components of Natural Remanent Magnetization (NRM). Heating to ~180°C removes a remanent magnetization parallel to the present-day field, which is interpreted as a viscous overprint. An intermediate unblocking temperature component is removed by heating to 450°C; this direction is always of normal polarity and is identical to a regional overprint imparted during the Early Jurassic emplacement of the Karoo Large Igneous Province. A high-temperature component isolated above 450°C is of dual polarity and is interpreted as primary on the basis of a positive reversals test. The virtual geomagnetic pole position for the Abrahamskraal Formation computed from the average high-temperature characteristic remanent magnetization direction is in agreement with the late Permian directions for stable Gondwana and with previous results from the lowermost Abrahamskraal Formation and Waterford Formation at the Ouberg Pass section. The predominantly normal polarity of this magnetization is in agreement with either a middle-late Lopingian age (ca. 254–256 Ma) or a late Guadalupian age (ca. 262 Ma) according to the global geomagnetic polarity time scale. We integrate these new results with existing magnetostratigraphic, biostratigraphic, and geochronological results from the Karoo Basin, with particular emphasis on the controversy over zircon age data reported from the underlying Ecca Group.

1. Introduction

The Paleo-Pacific or Panthalassan margin of Gondwana is the locus of magmatism, deformation, and sedimentation, dating from the first Ediacaran-Cambrian assembly of the great southern mega-continent until its Jurassic breakup. The Cape Fold Belt and the adjacent foreland Karoo Basin of southern Africa occupy the central segment of this 13,000 km continental margin. Indeed, the recognition of the common record of punctuated sedimentation and orogeny between the Cape Fold Belt/Karoo Basin and the Sierra Australes/Chaco-Paraná Basin of South America provided key evidence for a pre-Atlantic fit of Africa and South America [*du Toit*, 1937]. In the late Paleozoic, deformation and orogenic uplift along this margin has been linked by some authors to sedimentary deposition into the retro-arc foreland basins, i.e., the Karoo Supergroup of the Karoo Basin [*Lock*, 1978; *Winter*, 1984; *de Wit and Ransome*, 1992]. The age of deformation in the Western Cape Fold Belt is constrained by new $^{40}Ar/^{39}Ar$ muscovite ages of 275–260 Ma [*Hansma et al.*, 2015]. These new age data provide a firm basis for tectonostratigraphic correlation with the foreland basin deposits, calling for better age resolution of the Karoo Supergroup itself.

Although the broad lithostratigraphy of the Karoo Supergroup is well established [e.g., *Smith et al.*, 1993; *Catuneanu et al.*, 2005], detailed correlations are complicated by the heterolithic nature of continental sedimentary rocks. For example, the dynamic fluvial systems that deposited the Beaufort Group strata generated laterally variable facies over local and regional scales [e.g., *Gastaldo et al.*, 2005; *Ratcliffe et al.*, 2015]. Furthermore, present chronostratigraphic constraints reflect a mixture of vertebrate biostratigraphy [e.g., *Rubidge*, 1995; *Ward et al.*, 2005], palynological zonation [e.g., *Barbolini*, 2014], U-Pb geochronology on volcanic ashbeds [e.g., *Fildani et al.*, 2007, 2009; *Lanci et al.*, 2013; *Rubidge et al.*, 2013; *Gastaldo et al.*, 2015], and

magnetostratigraphy [*De Kock and Kirschivink*, 2004; *Ward et al*., 2005; *Lanci et al*., 2013; *Gastaldo et al*., 2015]. There are shortcomings inherent to each of these approaches, as well as calibration problems created by the combination of chronostratigraphic techniques; for example, the geomagnetic reversal pattern for the late Permian global polarity time scale is not fully resolved, and the geochronologic age for the boundaries of the magnetochrons is ill-defined [*Steiner*, 2006; *Shen et al*., 2011; *Gradstein et al*., 2012; *Szurlies*, 2013]. In this contribution, we present new magnetostratigraphic data from the lower-middle Beaufort Group, and integrate these results with previous magnetostratigraphic and existing chronostratigraphic data in order to address contradictory ages reported for the Ecca Group and overlying Beaufort Group. In particular, we examine the existing SHRIMP U-Pb zircon data from ashbeds in the Ecca Group to assess the validity of the latest Permian age interpretation for those rocks [*Fildani et al*., 2009; *McKay et al*., 2015].

2. Regional Geology

The Karoo Basin lies north of the Cape Fold Belt, and the Karoo Supergroup comprises the erosional detritus shed from the uplifting Cape Fold Belt [*Catuneanu et al*., 1998]. The Karoo Supergroup itself overlies the early-to-middle Paleozoic Cape Supergroup, principally composed of quartzose sandstones and subordinate shale that were deformed and metamorphosed during the late Permian Cape Orogeny [*Hälbich et al*., 1983; *Hansma et al*., 2015]. The base of the Karoo Supergroup is defined by glacio-marine diamictites of the Permian-Carboniferous Dwyka Group, which is overlain by deep water turbidites to fluvio-deltaic sediments of the Ecca Group. The Beaufort Group conformably overlies the Ecca Group, and is dominated by fluvial sandstones and mudstones [e.g., *Smith et al*., 1993; *Catuneanu et al*., 2005]. The base of the Beaufort Group marks the sedimentologic shift from deep water, marine, and deltaic facies to fluvial systems with episodic subaerial exposure, marked in the field by the lowermost occurrences of reddish, oxidized mudstones. The Abrahamskraal Formation is the lowest unit of the Beaufort Group, and this formation comprises a $\sim$2 km thick sequence of mudstones and sandstones [*Day and Rubidge*, 2014].

The lithostratigraphic divisions of the Karoo Group are well-defined, but the absolute age information is limited and some degree of diachronism is expected. Biostratigraphic ages for the Karoo Supergroup are based upon palynologic and faunal vertebrate records. The Ecca Group hosts a Early to Middle Permian palynologic assemblage [*Anderson*, 1977; *Falcon*, 1989; *Barbolini*, 2014; *Ruckwied et al*., 2014], with the occurrence of *Mesosaurus* fossils in the distinctive Whitehill Formation [*von Huene*, 1940] in the lower Ecca Group establishing a clear tie-point to the ca. 278 Ma Iratí Formation of the Passa Dois Group in the Paraná Basin of South America [*Santos et al*., 2006]. The contact between the Ecca Group and the Beaufort Group is defined as the boundary between the *Eodicynodon* and *Tapinocephalus* assemblage zones [*Rubidge et al*., 1999]. The Abrahamskraal Formation lies within the *Tapinocephalus* zone [*van der Walt et al*., 2010; *Day and Rubidge*, 2014; *Jirah and Rubidge*, 2014]. The upper Beaufort Group is interpreted to preserve a record of the Permian-Triassic mass extinction, which is represented by the boundary between *Dyconodon* and *Lystrosaurus* zones [e.g., *Maxwell*, 1992; *Botha and Smith*, 2007; cf. *Gastaldo et al*., 2015].

To complement this biostratigraphic database, there is a growing body of U-Pb zircon ages from volcanic ashbeds that are preserved in the Karoo Supergroup. The Dwyka Group is considered to be about 302–290 Ma in age based on ID-TIMS analysis of zircon [*Bangert et al*., 1999]. Studies of the overlying Ecca Group were carried out using the U-Pb SHRIMP method on zircons from volcanic ashes [*Fildani et al*., 2007, 2009; *McKay et al*., 2015]. These authors jointly interpret an age range of 252–276 Ma for deposition of the Ecca Group. However, the younger portion of this age range contradicts the 262–268 Ma age determined for the uppermost Ecca Group and basal Beaufort Group by *Lanci et al*. [2013] using U-Pb SHRIMP zircon geochronology paired with magnetostratigraphy. These latter authors report the 262–268 Ma age range as consistent with the presence of several normal polarity magnetozones, indicating an age younger than the Permo-Carboniferous Reverse Superchron (also known as the Kiaman Superchron). The ID-TIMS analysis of the middle Beaufort Group revealed U-Pb ages of 255–268 Ma using the chemical abrasion ID-TIMS method on zircon [*Rubidge et al*., 2013]. Notably, these last authors suggested a minimum age of 261.24 Ma for the top of the *Tapinocephalus* zone.

3. Sampling Locality Description

The studied sections are located at three closely spaced localities north of the town of Laingsburg along the Buffels River (Figure 1). The three sections comprise the entire stratigraphic record exposed by an

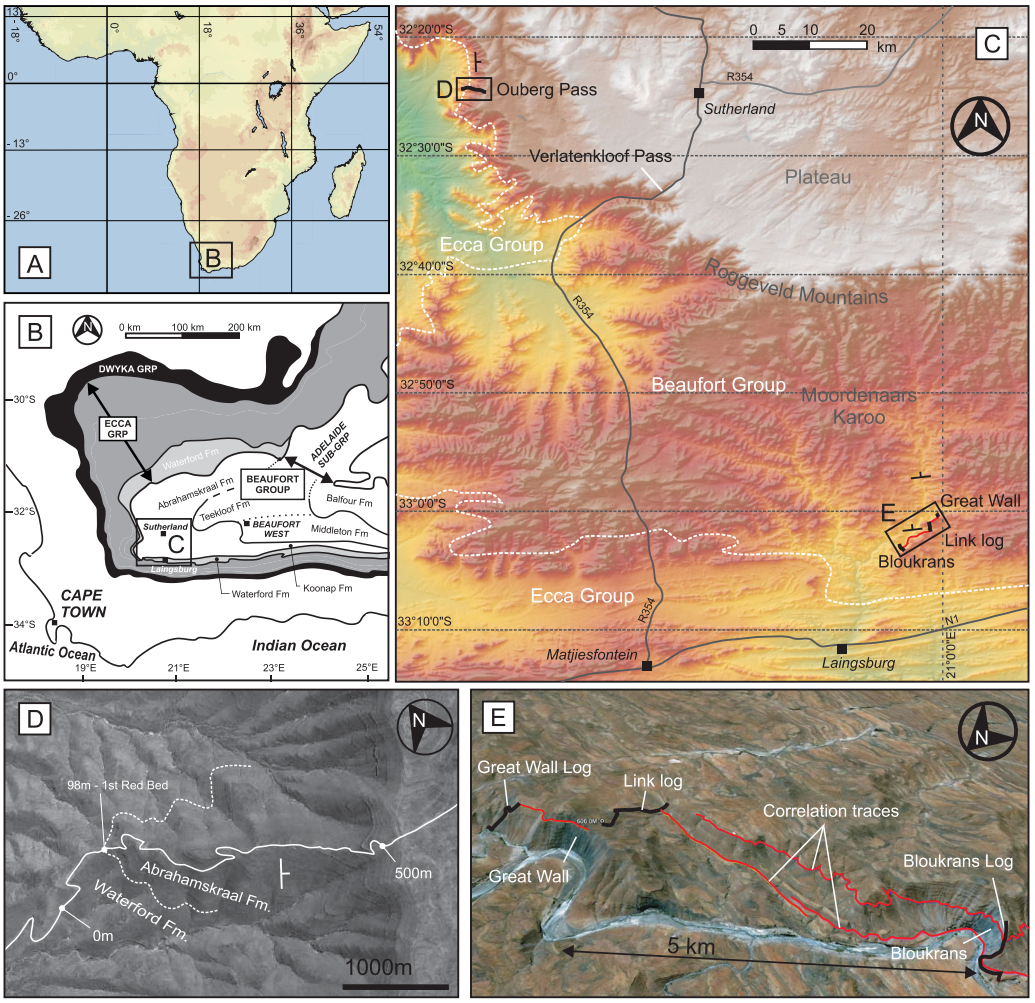


Figure 1. Study area and geology of the Karoo Basin. (a) Map of southern Africa showing the location of the southwestern Karoo Basin shown in Figure 1b. (b) Geological map of the southwestern Karoo Basin modified from *Johnson et al.* [1997]. The study area is shown as box C. (c) Topographic map of the study area. Logged sections are shown as solid black lines. Red lines are correlation lines linking the three logged sections in this study. White dashed line marks the approximate contact between the Ecca Group and the Beaufort Group in the study area. (d) Enlargement of the Ouberg Pass (OP) log, note the Ecca/Beaufort contact is at 96 m in the logged section. (e) Phototextured topographic model (exaggerated relief) showing the location of the Great Wall, Link, and Bloukrans logs (black lines) and correlation lines (red) between the sections (images from Google Earth).

open, NNE-SSW-oriented syncline that plunges shallowly to the E. The sections were measured, logged, and sampled from limb to core, i.e., from oldest to youngest. The lowermost "Great Wall" (GW) river cliff section (GPS coordinates 33°00.2′S, 20°59.4′E) crops out along the Buffels River (Figure 1) and can be correlated lithologically by walking out beds and using LiDAR data and Google Earth images (accessed February 2012) with the adjacent "Link" (LK) section (GPS coordinates 33°01.2′S, 20°58.8′E) which is in turn correlated to the uppermost "Bloukrans" (BK) section (GPS coordinates 33°02.9′S, 20°56.3′E) using aerial photos (Google Earth). The three sections altogether encompass the middle and upper part of the mudstone dominated fluvial Abrahamskraal Formation [*Wilson et al.*, 2014] for a composite thickness of about 600 m, with gap of about 5 m between the top of the GW section and the bottom of the LK section.

Evidence of the pervasive tectonic strain affecting the Cape Belt a few km south is shown at the studied locality by gentle folding, but pervasive rock fabrics are absent [*Hälbich*, 1994]. A poorly developed cleavage is apparent from flat joint surfaces decorated by submillimeter thick, bladed calcite, which we interpret as parting along an incipient cleavage fabric. This incipient cleavage has an E-W orientation parallel to regional fold-axes. Bedding is subhorizontal at GW ranging from 035/05 (dip direction/plunge) to 220°/05°; it ranges

from 210°/28° to 210°/30° at LK and is 145°/13° at BK. The base of the GW-LK-BK section was mapped at 340 m stratigraphically above the first red bed occurring along the Buffels River north of Laingsburg. All three sections expose similar lithologies and, as discussed later, magnetic properties. Standard 2.5 cm diameter paleomagnetic samples were taken from all exposed beds at an average stratigraphic interval of ~2 m using a gasoline-powered drill. Samples were oriented using magnetic and solar compasses. Bulk magnetic susceptibility was measured in the field using a hand-held SM20 magnetic susceptibility meter manufactured by GF Instruments.

4. Results

4.1. Magnetic Susceptibility and Tectonic Fabric

The bulk magnetic susceptibility measurement shows an aperiodic variation of SI values in the 10^2-10^4 range, with the highest values associated with cm-scale, black horizons in coarse-grained sandstone beds. Electron dispersive spectroscopy of polished thin sections of these horizons indicates the predominance of low-Ca, low-Mn almandine-pyrope garnets, and ilmenite, suggestive of a basement source region that contains amphibolite to granulite-facies pelitic or mafic rocks [*Arosio*, 2012]. In contrast, the highest bulk susceptibility values from the Ouberg Pass never exceeded 10^1 SI units, indicating increasing values of magnetic susceptibility up-section for the Beaufort Group, with implications for evolving source area(s). The presence of placer deposits of heavy minerals derived from metamorphic basement suggest that a significantly exhumed basement source region contributed to the detrital input, in addition to erosional recycling of the low-grade metasedimentary rocks that form the Cape Supergroup.

The orientation of the tensor of Anisotropy of Magnetic Susceptibility (AMS) was measured on a Kappabridge KLY-3 susceptibility meter using the standard 15 orientation protocol. The AMS is characterized by well-developed anisotropy of both oblate and prolate ellpsoids. The axis of maximum susceptibility (K_1) is tightly clustered in a nearly horizontal, E-W direction, and the intermediate (K_2) and minimum (K_3) axes form a girdle of N-S, horizontal to vertical positions (Figure 2). The clustering of K_1 axes might result from the preferential orientation of elongate minerals due to pervasive deformation. However, the absence of a lineated macroscopic fabric suggests a different explanation, where the K_1 axis reflects the intersection of horizontal bedding and a vertical foliation [*Parés et al*., 1999]. This AMS pattern is typical of weakly developed cleavage in mudstones, formed by the same deformation event that created the synclinal structure of the GW-LL-BK site. In contrast, the AMS pattern reported by *Lanci et al*. [2013] from the Ouberg Pass section, ~100 km NW of the GW-LL-BK site, is typical of sedimentary fabrics, i.e., a uniformly oblate fabric with K_3 axes in a steep orientation perpendicular to bedding, and (K_1) axes dipping shallowly to the S, suggestive of imbrication caused by N-directed paleocurrents. The preservation of purely sedimentary fabrics at Ouberg Pass and partial tectonic overprinting at the GW-LL-BK site defines a strain gradient that decreases to the N, consistent with the relative proximity of these sites to the foliated and deformed rocks of the Cape Fold Belt.

4.2. Characteristic Remanent Magnetization (ChRM) and Polarity Sequence

The Natural Remanent Magnetization (NRM) was studied in a set of pilot samples by stepwise alternating field and thermal demagnetizations; thermal demagnetization yielded the most interpretable results and was used as standard procedure. Paleomagnetic directions were determined by least squares line-fitting using principal component analysis and avoiding the use of remagnetization circles. The inferred magnetic mineralogy of the studied sections is virtually identical to that of the Ouberg Pass section [*Lanci et al*., 2013], as confirmed by IRM acquisition on a set of pilot samples (Figure 3). Like the Ouberg Pass locality, the NRM comprises two distinguishable components, as well as a present-day field overprint, which is completely removed by heating to 150°C where present. A consistently observed paleomagnetic direction (component-A) is isolated by stepwise heating to temperatures between 150 and 450°C and AF peak field below 90 mT. Component-A is well grouped and features a single, normal polarity with a Fisher mean direction of about Dec. 344.3/Inc. −56.6. This mean direction is practically identical to the homonymous component from Ouberg Pass [*Lanci et al*., 2013], is isolated within the same range of temperatures, and is consistent with a well-known regional, secondary overprint [e.g., *Ballard et al*., 1986]. Component-A from this study and from *Lanci et al*. [2013] are plotted in Figure 4 for a direct comparison.

The higher laboratory unblocking temperature magnetization (component-B), which was interpreted as Characteristic Remanent Magnetization (ChRM), was isolated by thermal demagnetization at temperatures

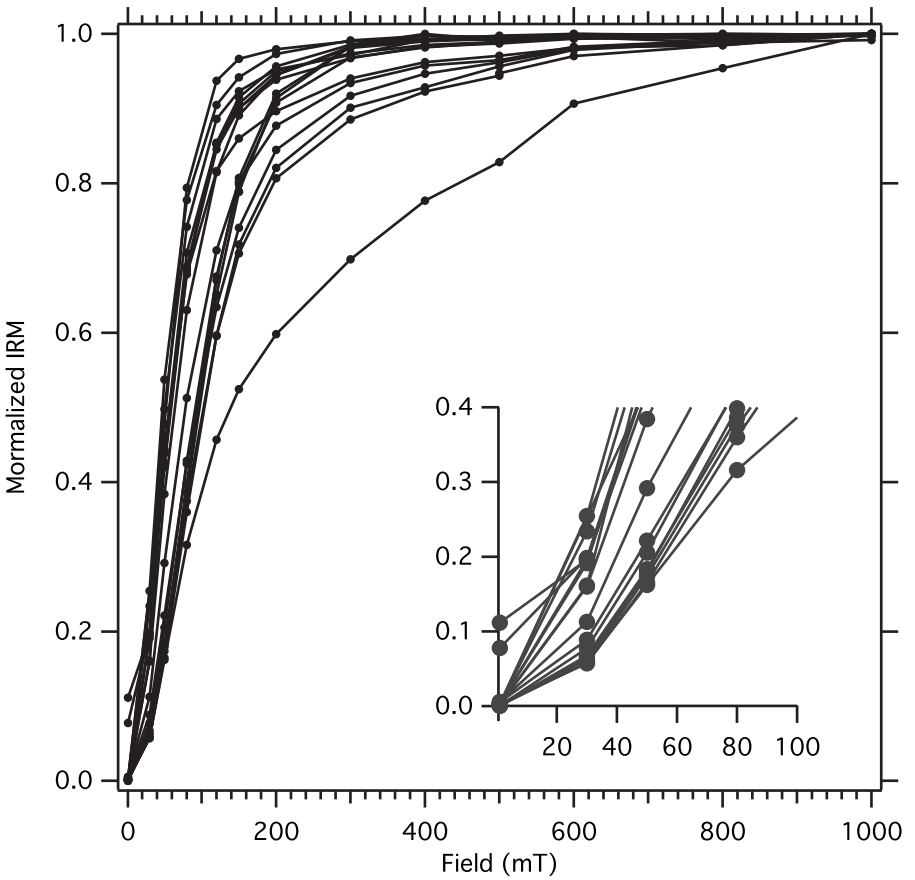


Figure 2. Pattern of principal axes of magnetic susceptibility in the Abrahamskraal Fm. at the Great Wall section. Jelinek and Flinn fabric diagrams and the equal-area plot (lower hemisphere) of the tilt-corrected directions of the principal axes of AMS (squares, K_1; triangles, K_2; circles K_3) are shown. The great circle represents the bedding planes in the adjacent Link and Bloukrans section; the bedding plane at the Great Wall section is horizontal. The major K_1 axis is well grouped in the East-West direction suggesting the presence of an intersection lineation between bedding and E-W trending cleavage formed during Cape Fold Belt deformation.

ranging from 450 to 580°C. A small number of samples, probably hematite-bearing, show maximum unblocking temperatures higher than 600°C. Notably, not all samples exhibit both components A and B; in particular, component-B was successfully isolated only in about one third of the measured samples (n = 98, N = 261). Most of the specimens that failed to show component-B were either completely demagnetized by heating to 450°C or became unstable at higher temperatures. Samples that preserve both the A-component and a normal polarity B-components are typically marked by an inflection in declination from a NNW to a NW-WNW direction, with little change in inclination. The inflection point commonly occurs at ~450°C of laboratory heating (e.g., samples LS27A in Figure 8 and sample BK32a in Figure 9). This inflection is much more obvious for samples that preserve the A-component and a reversed polarity B-component (e.g., samples GW40C and GW93A in Figure 7, sample BK42A in Figure 9). A group of about 40 samples mainly from the upper part of the BK section that were sampled up to the ridgeline (meters 190–300, stratigraphic height in BK) showed high intensities of magnetization and well-aligned, single-component NRMs. The directions of magnetization for these ridgeline samples appear to be random, with highly dispersed declinations and mostly shallow inclinations. This group of samples were interpreted as remagnetized by lightning and disregarded from further consideration.

Tilt-corrected paleomagnetic directions from component-B (Figure 5a) pass a B-class reversal test (critical angle $\gamma_c = 9.48°$; Table 1). The distributions of ChRM directions from Ouberg Pass and this study overlap when plotted on the same equal-area diagram (Figure 5b) and the joint set of ChRM directions shows a dual polarity, antipodal directions that pass a reversal test (Figure 6). Some improvement in the critical angle value for the compiled data set ($\gamma_c = 5.92°$) reflects the similar proportions of normal and reversed

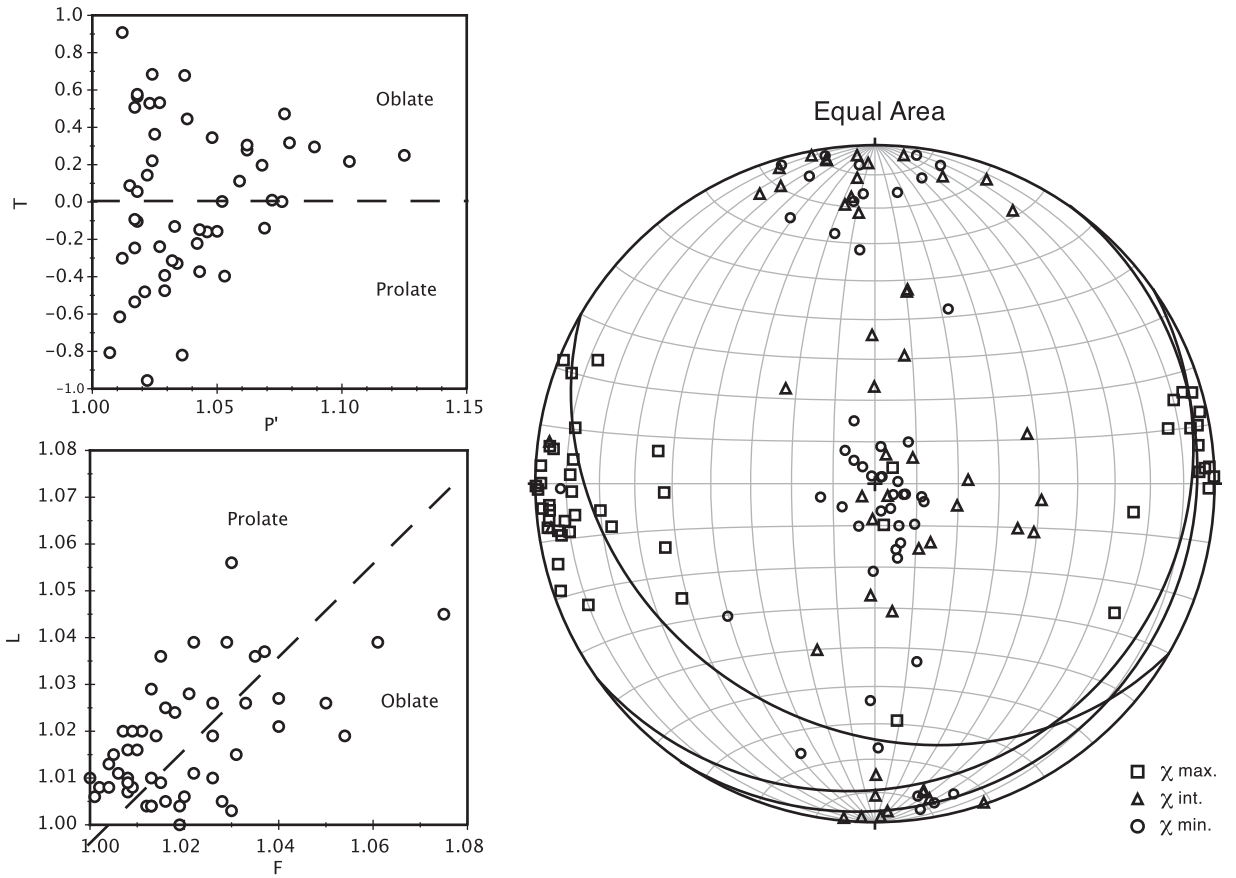


Figure 3. Normalized IRM acquisition curves of a representative set of samples from BK and LK sections. With a few exceptions samples are saturated at field <300 mT suggesting that the dominant magnetic mineral is magnetite, in agreement with coeval samples from Ouberg Pass section [*Lanci et al*., 2013]. Sporadic high-coercivity samples are found in the most oxidized bedding. The inset shows the high NRM of two samples remagnetized by lightning.

directions. ChRM directions (component-B) from this study are similar to the results from Ouberg pass (Table 2). The latitude of the virtual geomagnetic pole (VGP Lat.) are plotted in Figures 7–9 for the GW, LK, and BK sections, respectively, together with representative examples of orthogonal vector plots. In the resulting composite magnetic stratigraphy (Figure 9), which has a prevalence of normal polarity magnetizations, we identify three normal and three reverse polarity zones. Each polarity zone was defined on the basis of two or more stratigraphically consecutive samples with the same polarity.

5. Discussion

5.1. Stratigraphic Correlation and Magnetostratigraphic Considerations

The rock-magnetic properties, the direction of the magnetic overprint related to the Karoo Large Igneous Province, and the average ChRM direction from this study are all very similar to those reported in the uppermost Waterford Formation and lowermost Abrahamskraal Formation at the Ouberg Pass by *Lanci et al*. [2013]. However, the composite GW-LK-BK section has a predominance of normal geomagnetic polarity magnetozones, which contrasts with the dominantly reverse polarity of Wordian/Capitanian stages, the time interval of deposition of the uppermost Ecca Group and lowermost Abrahamskraal Fm. at Ouberg Pass [*Lanci et al*., 2013]. No ashbeds were identified in the GW-LK-BK section, so there is no geochronologic data that we can use to better define the age of sedimentation and acquisition of a ChRM. In the following discussion, we attempt to place these new magnetostratigraphic results into the context of both the stratigraphy of the Karoo Basin, current knowledge of the global geomagnetic polarity time scale, and existing geochronologic data sets that serve to define the upper and lower bounds of age of the GW-LK-BK section.

The geochronologic and magnetostratigraphic data sets reported by *Lanci et al*. [2013] form the first constraint on the age of the GW-LK-BK section. A physical correlation of strata between these two localities is

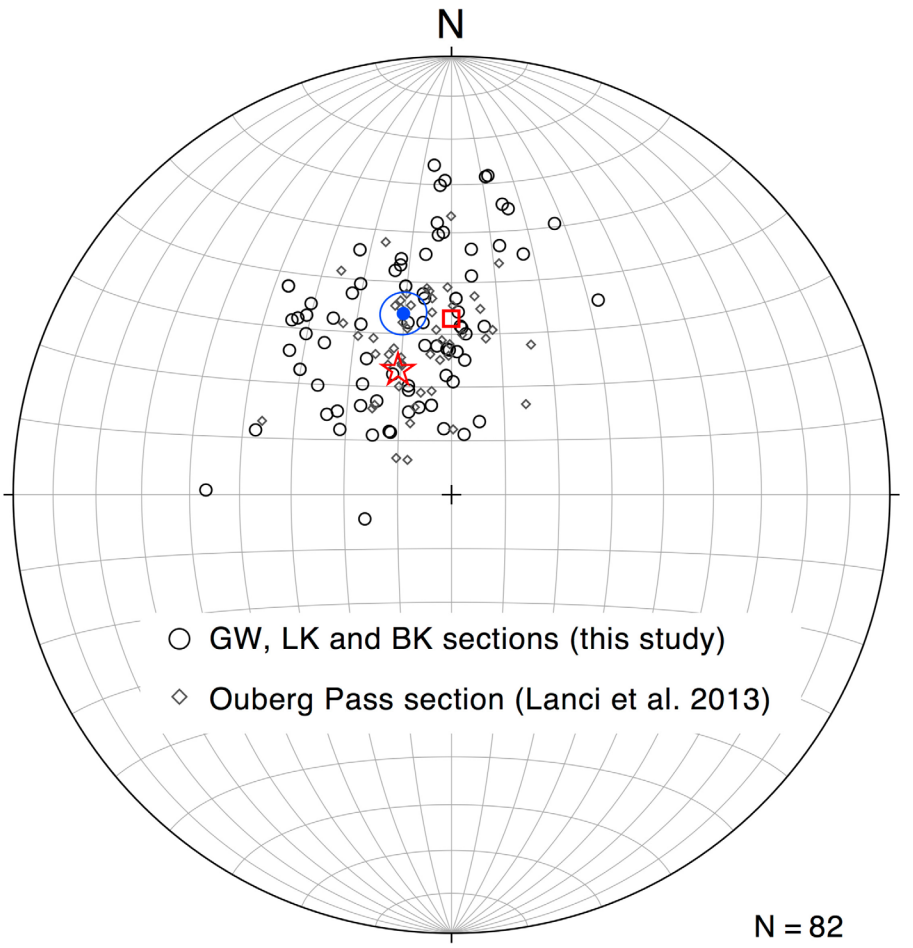


Figure 4. Equal-area stereoplot (lower hemisphere) of the directions of component-A in geographic coordinates from the GW, LK, and BK sections (circles), and from the Ouberg Pass sections (diamonds) showing the perfect overlap in the two sites. Averaged direction and 95% confidence cone are shown. Red square shows position of time-averaged geomagnetic field direction, and red star shows present-day geomagnetic field direction.

impossible because of the ~100 km distance between the section localities, and the degree to which sedimentation rates vary over this distance is unknown. Our use of the contact between the Ecca Group (Waterford Fm.) and the Beaufort Group (Abrahamskraal Fm.) as a first-order datum (marked by a dashed white line in Figure 1c) places the GW-LK-BK sections stratigraphically higher than the Ouberg Pass section. The stratigraphic thickness between the base of the GW section and the top of the underlying Waterford Formation was calculated based on map relations to be ~340 m. The lowermost Abrahamskraal Formation in the Ouberg Pass section is dominated by reversed polarity magnetozones, with short, normal polarity magnetozones interpreted to represent some part of the lower part of the Illawarra series, which post-dates the Permo-Carboniferous Kiaman Reversed Superchron [*Lanci et al.*, 2013]. This age assignment for the Ouberg Pass section is in agreement with the 262–268 Ma range in ages of volcanic ashbeds found in this ~600 m thick section [*Lanci et al.*, 2013], and the ca. 260 Ma ID-TIMS zircon age determined for the upper Koonap Fm., the lateral equivalent of the Abrahamskraal Fm ~500 km east of our study area [*Rubidge et al.*, 2013].

Following the stratigraphy outlined by *Catuneanu et al.* [2005], we present two possible interpretations for the magnetozones observed in the GW-LK-BK section. One possible interpretation correlates the normal magnetozones of the GW-LK-BK sections with the N1–N3 "Capitanian" normal zone (ca. 262–264 Ma), and indicates an equivalent age of the GW-LK-BK section with the uppermost portion of the Ouberg Pass section (Figure 10, orange shading). However, the greater thickness of the Capitanian normal zone in the GW-LK-BK section implies a sedimentation rate about 1 order of magnitude higher than that deduced for the Ouberg Pass section (~70 m/Ma). Such a change in sedimentation rates may be feasible for fluvial environments associated with mega-fan architecture described for the study area by *Wilson et al.* [2014] and *Ratcliffe et al.*

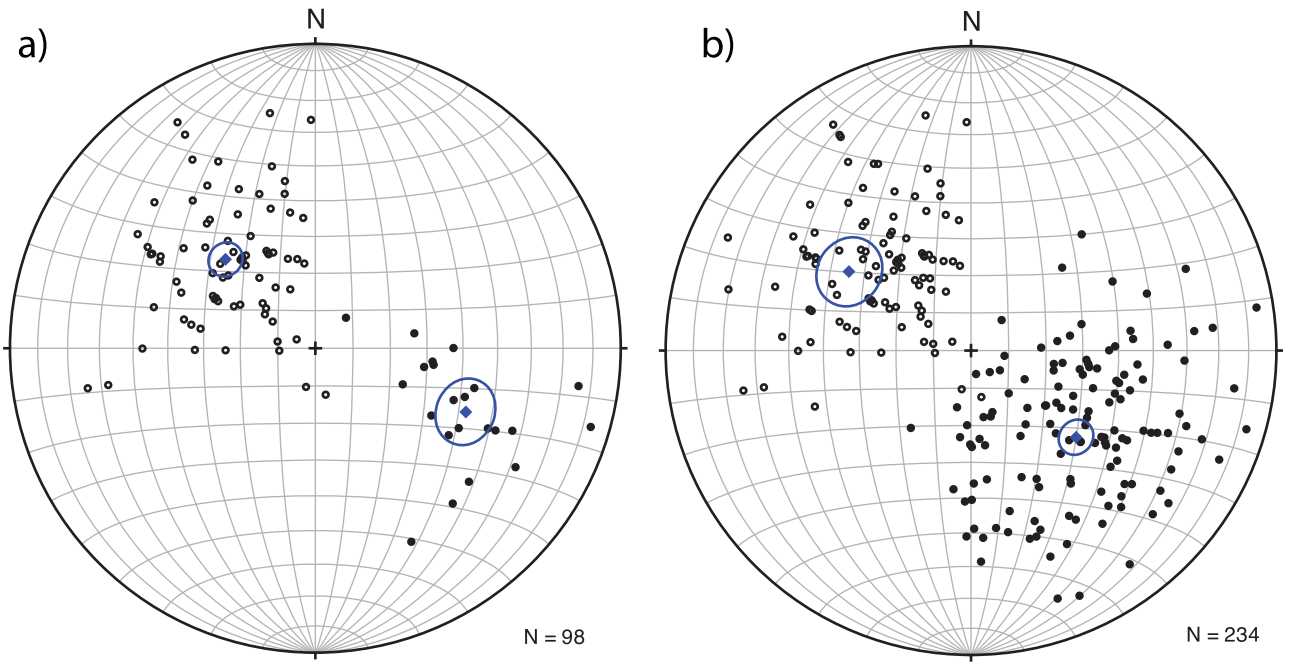


Figure 5. (a) Equal-area stereoplot (lower hemisphere) of the tilt-corrected ChRM directions (component-B) from the GW, LK, and BK sections and form (b) the joint set of Ouberg Pass and the GW, LK, and BK sections. Averaged directions and 95% confidence cone are shown for each separate polarity. Results from Ouberg Pass and this study overlap perfectly and normal and reversed directions are antipodal and pass a reversal test. Detailed statistics are reported in Table 1.

[2015]. The second interpretation involves correlating the N1–N3 normal polarity intervals to the uppermost part of the Illawarra mixed-polarity zone (Figure 10, green shading), i.e., the middle Wuchiapingian to early Changshingian (ca. 254–257 Ma).

5.2. Inclination Shallowing and Pole Position

We applied the Elongation-Inclination methods of *Tauxe and Kent* [2004] to examine the GW-LK-BK rocks for the effects of sediment compaction and attending inclination shallowing. Unlike the results from the

Table 1. Reversal Test Statistics

						Eigen Values				
Great Wall-Link-Bloukrans (GLB)	Dec	Inc	R	k	alpha(95)	E3	E2	E1	T2 conf(95)	T1 conf(95)
Reversed Directions, N = 22										
Fisher statistics	113.0	45.5	20.56	14.62	8.4					
Bingham statistics	113.0	45.7				19.3	1.92	0.74	0.92	0.89
Normal Directions, N = 76										
Fisher statistics	314.8	−55.9	70.58	13.84	4.5					
Bingham statistics	315	56.1				66.03	5.89	4.08	0.57	0.56
Angle between Fisher averages				17						
Angle between Bingham averages				17.21						
Critical angle [*McFadden and McElhinny*, 1990]				9.48						

						Eigen Values				
Compiled GLB + OP (Ouberg Pass)	Dec	Inc	R	k	alpha(95)	E3	E2	E1	T2 conf(95)	T1 conf(95)
Reversed Directions, N = 135										
Fisher statistics	126.5	52.1	120.96	9.55	4.2					
Bingham statistics	126.7	52.4				109.52	14.32	11.17	0.56	0.55
Normal Directions, N = 99										
Fisher statistics	311.8	−54.8	91.62	13.27	4.1					
Bingham statistics	312.1	−55.2				85.41	7.38	6.21	0.54	0.53
Angle between Fisher averages				4						
Angle between Bingham averages				4.29						
Critical angle [*McFadden and McElhinny*, 1990]				5.92						

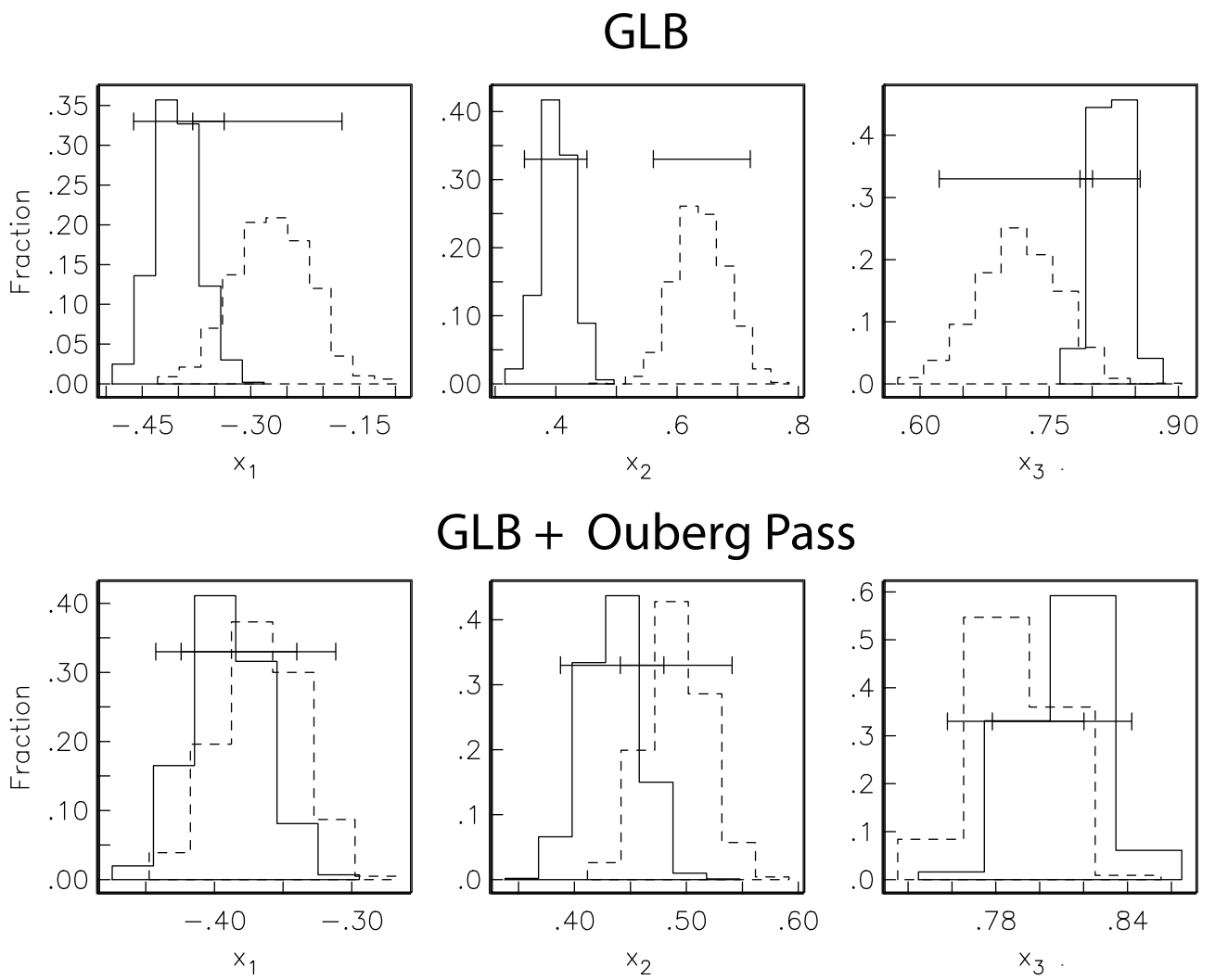


Figure 6. (top) Bins of common polarity bootstrapped results for the Great Wall-Link-Bloukrans (GLB) ChRM component-B directions in Cartesian coordinates (dashed line, reversed directions; solid line, normal directions). (bottom) Compiled directions from GLB and Ouberg Pass (OP) demonstrating common mean at 95% confidence level.

Ouberg Pass section [*Lanci et al*., 2013], the composite BK-LK-GW data set does not exhibit significant inclination shallowing for the B paleomagnetic direction despite the fact that the average direction and pole position is virtually identical to those of Ouberg Pass. Conversely, the same analysis on the entire set of ChRM directions from *Lanci et al*. [2013] and this study, gave again an E-I corrected inclination of −62.9° with a relatively large confidence interval and a flattening of about 9.6° (Table 2), possibly due to the influence of the Ouberg Pass data on the overall distribution. No significant inclination shallowing was identified using the E-I technique on the A-component directions from the joint set of *Lanci et al*. [2013], as expected given its origin as post-compaction, thermal overprint. Whether or not the inferred inclination shallowing is a real feature of the paleomagnetic record cannot be established, but the E-I uncorrected average pole position for the whole set of samples from the Abrahamskraal and Waterford Formations reported in Table 2, is consistent with the pole position reported by *De Kock and Kirschivink* [2004] for sediments that potentially span the Permian-Triassic boundary.

5.3. U-Pb Geochronological Data and Age Controversy

Either of the two correlations proposed for the new magnetostratigraphic data are consistent with other geochronologic data from the Beaufort Group [*Lanci et al*., 2013; *Rubidge et al*., 2013; *Gastaldo et al*., 2015]. There exists an outstanding controversy regarding the age of the underlying Ecca Group, based on the geochronologic results presented by *Fildani et al*. [2007, 2009] and recently extended by *McKay et al*. [2015].

Table 2. Average Paleomagnetic Directions (Fisher), the Virtual Geomagnetic Pole Position is Computed From the Fisher-Mean Direction

	Dec	Inc	k	α_{95}	N	VGP, Lat. (dp)	VGP, Long. (dm)	VGP, Lat.[c]	VGP, Long.[c]
A-component	344.3	−56.6	17.7	3.8	82	76.5 (4.0)	268.6 (5.5)		
ChRM (B-component) normal	314.8	−55.9	13.8	4.5	76				
ChRM (B-component) reversed	113.0	45.5	14.6	8.4	22				
ChRM (B-component) combined	309.0	−53.9	12.8	4.2	98	48.0 (4.1)	274.4 (5.9)	55.7	272.6
ChRM (B-component)[a] [*Lanci et al*., 2013, this study]	308.7	−53.3	10.8	2.9	234	47.5 (2.8)	274.0 (4.0)	55.2	272.2
E-I corrected inclination[a]		−62.9		56–70[b]	234	48.9 (8.6)	258.8 (11.0)	56.1	254.1
Domeier et al. [2011b] (volcanic Argentina)								52.4	244.3

[a]Site coordinate: Lat. = −32.7, Long. = 20.6.
[b]Bootstrapped 95% confidence interval.
[c]Rotated to northwest Africa (Adria) coordinates using rotation parameters of *Lottes and Rowley* [1990].

The former two contributions reported U/Pb SHRIMP zircon ages ranging from 274.8 to 252.7 Ma for ash deposits in the Laingsburg Formation (Ecca Group), which underlies the Abrahamskraal Formation near our study area. Taken at face value, the *Fildani et al*. [2007, 2009] results would place the upper Ecca Group in the latest Permian, thus implying a Triassic age for the overlying Abrahamskraal Formation. This age assignment was reinforced by *McKay et al*. [2015], who reported additional U-Pb SHRIMP zircon data from ashbeds in the Ecca Group rocks of the Tanqua, Central Basin, and Ripon depocentres. *McKay et al*. [2015] propose that the "inverted" age stratigraphy represents an early Triassic period of zircon exhaustion in the putative source for these ashbeds, the Choiyoi magmatic province of Argentina, pointing to the presence of recycled zircon in the ca. 245 Ma Puesto Viejo Formation studied by *Domeier et al*. [2011a,b].

There are three substantial problems with the 250–255 Ma age interpretation for the upper Ecca Group strata. First, assigning the Ecca Group to the latest Permian ignores the existing biostratigraphic observations presented by faunal [*Rubidge*, 1990; *Smith and Keyser*, 1995; *Rubidge et al*., 1999] and palynologic data [*Barbolini*, 2014], which uniformly support correlation with middle Permian rocks of the other Gondwanan continents. Because most period or era boundaries are fundamentally biostratigraphic divisions, e.g., the Permian-Triassic boundary separating Paleozoic and Mesozoic biota, abandoning these constraints ignores an enormous, coherent body of chronostratigraphically vital data. A second contradiction is posed by the identification of post-Kiaman magnetic polarity zones and associated 262–268 Ma SHRIMP zircon ages reported by *Lanci et al*. [2013] from the basal Abrahamskraal Fm., consistent with CA-TIMS U-Pb zircon ages of 255–268 Ma reported by for the lower and middle Beaufort Group [*Rubidge et al*., 2013], and a new ca. 253 Ma age for a volcanic porcellanite in the uppermost Beaufort Group reported by *Gastaldo et al*. [2015]. Though the "zircon exhaustion" hypothesis could explain the problem of apparently inverted stratigraphy, there is no documentation at present as to the true source of the Karoo ash deposits. The presence of inherited zircons in the Puesto Viejo Fm. of Argentina [*Domeier et al*., 2011b] does not prove that all ash deposits of late Permian age in the Karoo Basin contain exclusively inherited zircon. The third contradiction stems from the association between the Cape Fold Belt and foreland basin sediments of the Karoo Supergroup. Extensive faulting and folding has affected the Ecca Group in the Laingsburg region, which is situated in the fold-thrust portion of the Cape Fold Belt. New $^{40}Ar/^{39}Ar$ data reported by *Hansma et al*. [2015] from shear zones along strike from Laingsburg indicate that this section of the fold-thrust belt was active between 260 and 275 Ma. This range of deformation ages provides a minimum constraint of ca. 260 Ma for the deposition of the Ecca Group strata.

Because this disagreement arises solely from different interpretations of U-Pb SHRIMP geochronologic data, we consider that the problem may be resolved by uniform scrutiny of the U-Pb zircon data. *Lanci et al*. [2013] argued that the "young" age interpretations reported by *Fildani et al*. [2009] could reflect undetected Pb-loss from a relatively small number of zircon grains, a problem compounded by the use of the single youngest, concordant zircon grain (or grains, in some cases) as the basis of the age interpretation. This selection implies that all older zircons are inherited xenocrysts, which introduces a young bias to the age interpretation. We have re-examined the U-Pb zircon data set reported by *Fildani et al*. [2007, 2009] to explore alternative age interpretations that might mitigate the disagreement. Specifically, we have employed a more conservative criterion that assigns an age to the youngest population of

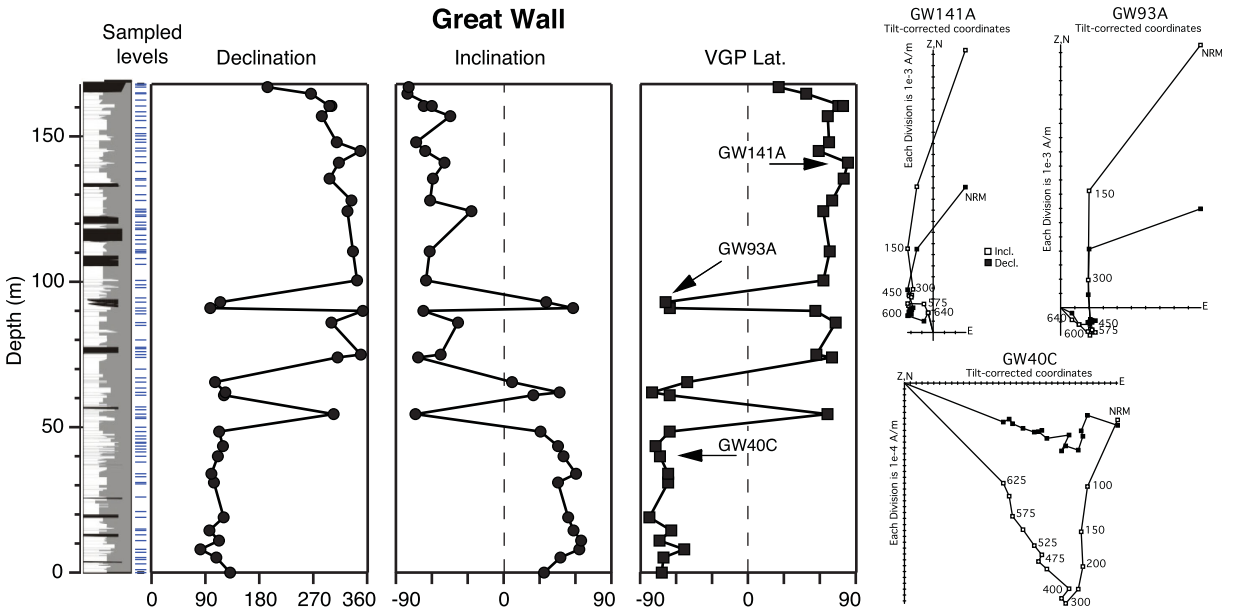


Figure 7. Summary of magnetic stratigraphy of the Great Wall section with lithological description, sampled levels, ChRM paleomagnetic directions, virtual geomagnetic pole latitude (VGP Lat), magnetic susceptibility, and representative orthogonal vector plots are reported. In the stratigraphic column mudstones are shown with a white fill and sandstones with a black fill, background is gray.

zircons, which we interpret as the eruption age of the ashbed. Notably, the precision of secondary ions mass spectrometry isotopic data for ages $<$500 Ma does not commonly allow for discordance to be detected, only suspected. The loss of Pb from a zircon grain results in migration of the analytical result toward the origin in U-Pb space. For ages of ca. 250 Ma, the Pb-loss discordant array is parallel to the concordia curve, so Pb-loss can occur without being detected. This creates the quandary of analytically

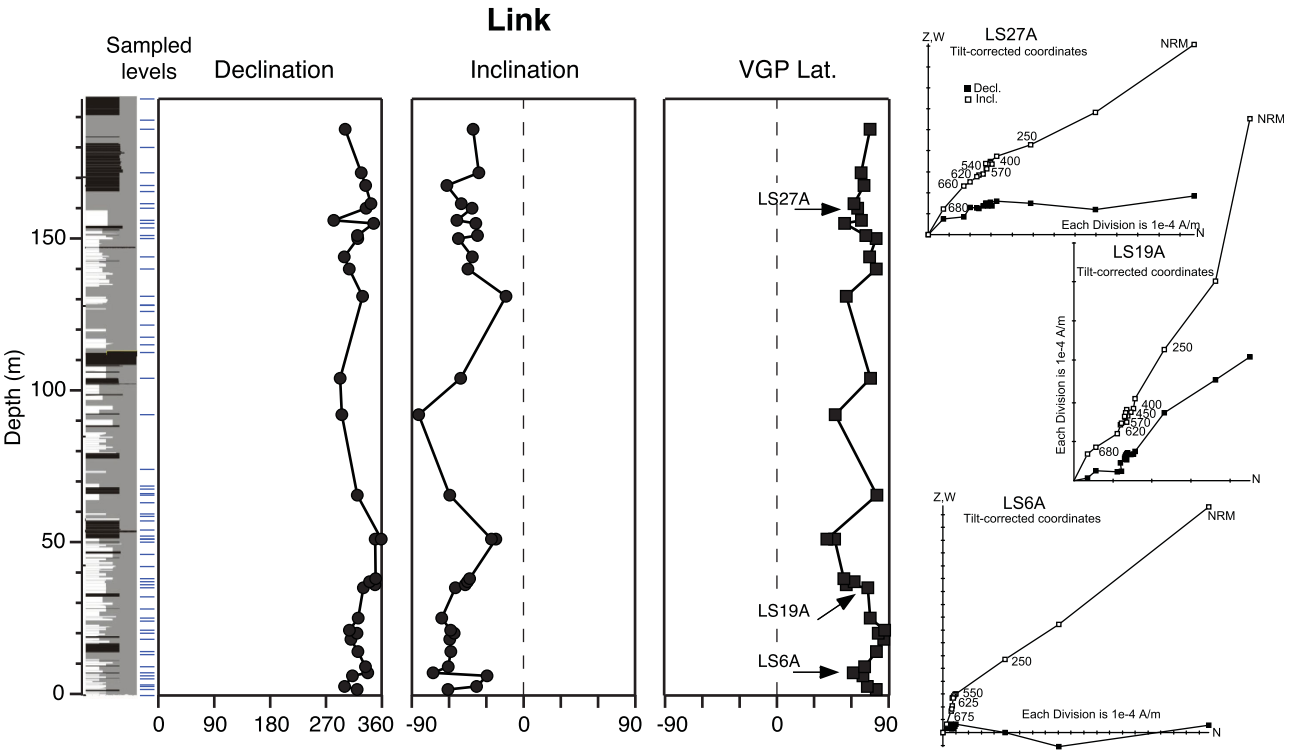


Figure 8. Summary of magnetic stratigraphy of the Link section, with stratigraphic log, sampled levels, ChRm directions, magnetic polarity, and selected Zijderveld plots showing demagnetization behavior. Colors in stratigraphic column as in Figure 7.

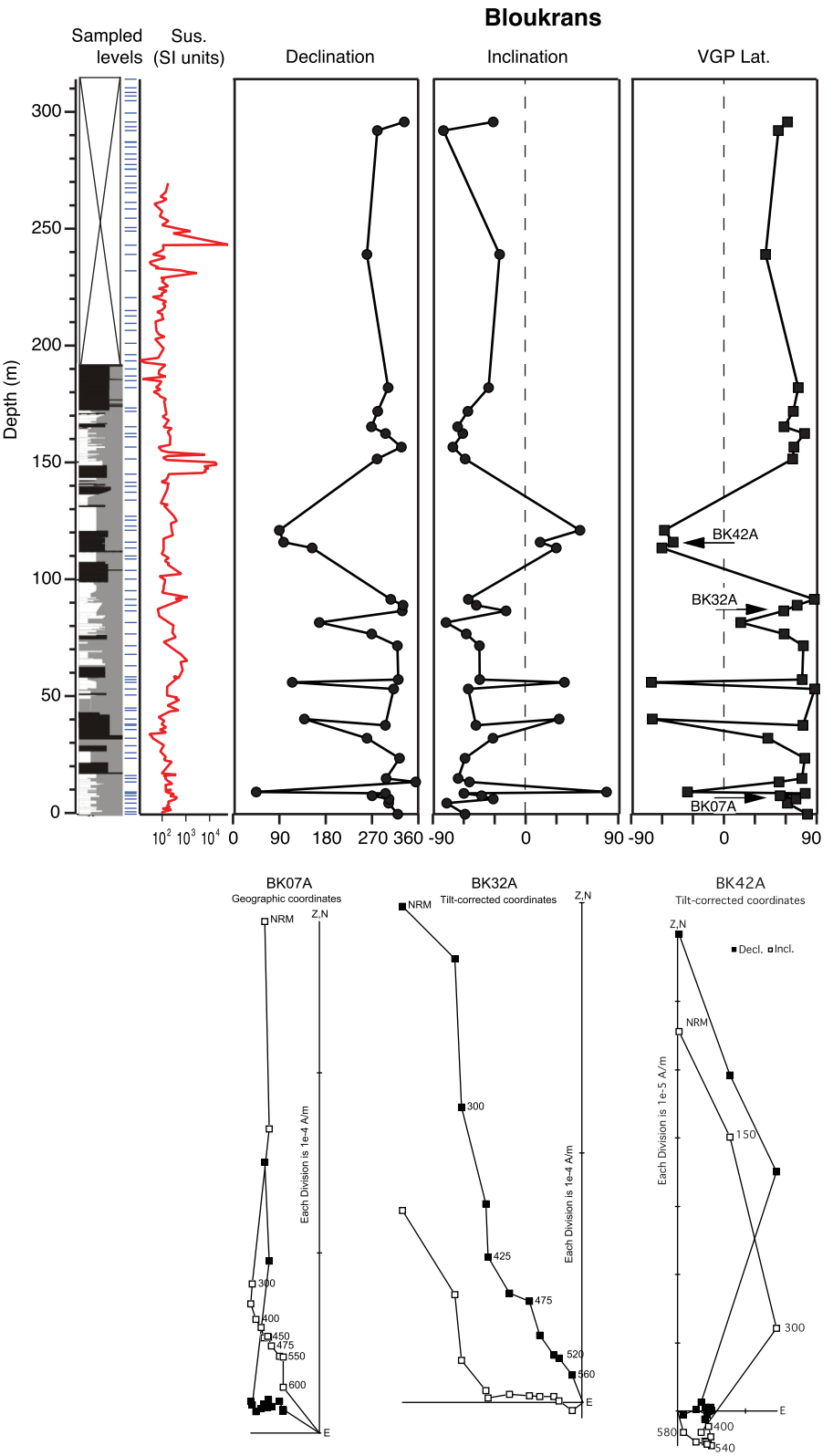


Figure 9. Summary of magnetic stratigraphy of the Bloukrans section, with stratigraphic log, sampled levels, magnetic susceptibility, ChRm directions, magnetic polarity, and selected Zijderveld plots showing demagnetization behavior. Colors in stratigraphic column as in Figure 7.

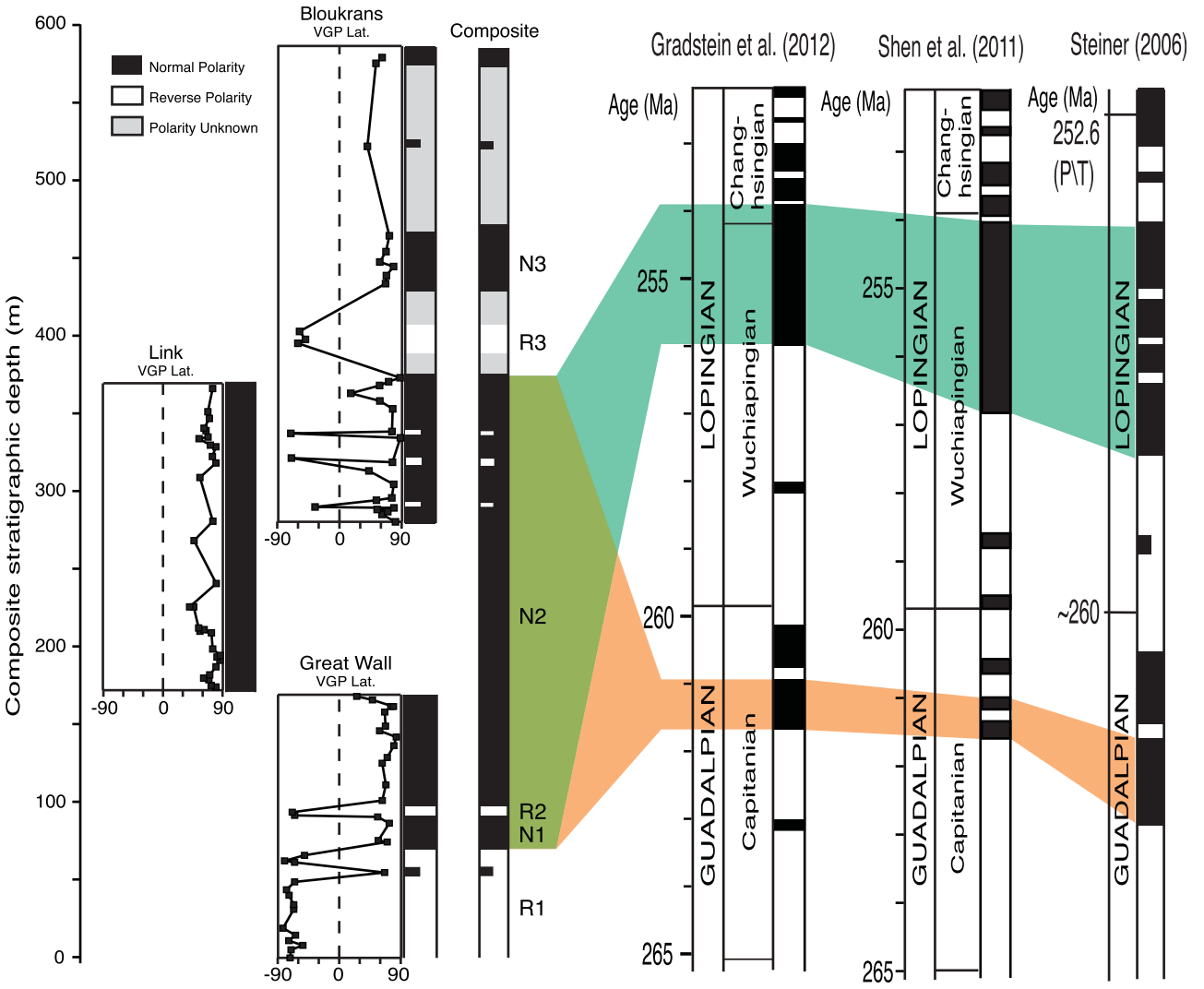


Figure 10. Summary of the magnetic stratigraphy of the joint GW, LK, and BK sections compared to recent geomagnetic polarity time scales, indicating both of the possible correlations.

concordant data that are not marked by inordinately high common Pb, so the initial suspicion of undetected Pb-loss is based solely on the fact that such grains are outliers to the main population. Fortunately, a reasonably large amount of analytical data from different zircon grains is easily obtainable with the SHRIMP, so that coherent populations can be recognized. Where undetected Pb loss is suspected, the grain(s) so affected should be younger than the youngest unaffected zircon population.

In considering the age of the Ecca Group ashbeds, we restrict our re-interpretation of ages to those data reported by *Fildani et al*. [2009] from the ashbeds in the Laingsburg region, avoiding the uncertain correlation between separate submarine fan deposits of the Lainsburg and Tanqua regions, which are ~100 km apart. This caution is warranted by reported diachroneity in the age of the contact between Ecca Group and Beaufort Group strata [*Turner*, 1999; *Rubidge et al*., 1999]. Similar reasoning leads us to exclude the Ripon depocentre data recently presented by *McKay et al*. [2015] as well, although the arguments made below pertain to that data set as well. We use the TuffZirc algorithm [*Ludwig*, 2009] to identify zircon populations that we interpret as the best geochronologic estimate of an eruption age (Table 2). Results from the TuffZirc analysis yield ages in the 265–275 Ma range (Figure 11), which is in reasonable agreement with the 262–268 Ma age assignment for the uppermost Ecca Group and lowermost Beaufort Group exposed at Ouberg Pass [*Lanci et al*., 2013]. The notable exception to this 265–275 Ma age range is sample CVX-12, which presents a TuffZirc

Table 3. Reinterpreted of Ecca Grp Data From *Fildani et al.* [2009]

Ecca Group Laingsburg Region Sample Name	*Fildani et al.* [2009] Interpreted Age Result			TuffZirc Identifed Population				Monte Carlo Stratigraphic Age of TuffZirc Populations	
	Age	Error (2σ)	# Grains	Age	Error	Conf. Level	# Grains	Age	Error ± (95%)
CVX12[a]	252.7	4	6	256	+2/−3	93	8	256.2	+1.8/−3.3
CVX11	254.2	6.4	1	265.5	+3.5/−0.5	96.1	12	265.8	+1/−1.7
CVX10[b]	258.1	3.2	1	266	+1/−1	93.8	5	266.56	+0.68/−1.2
CVX6	261.7	2.8	3	275	+1/−4	96.1	9	271	+2.1/−1.9
CVX8	262.8	4	9	266	+5/−3	97.8	13	271.9	+2.3/−2
CVX5	270.1	4	2	278	+4/−4	87.8	4	275.1	+1.5/−2.2
CVX4	274.8	3	4	275	+2/−1	96.1	12	276.3	+1.2/−1.8

[a]Suspect sample with anomalously young age.
[b]Tuff zirc identified an additional, older population of 276 +4/−1 Ma (n = 9, conf. = 96.1%).

age of 256 +2/−3 (n = 8, 93% conf.), somewhat older than the mean-weighted $^{206}Pb/^{238}U$ age of 252.7 ± 4 Ma (n = 6, 2σ error) reported by *Fildani et al.* [2009]. Although the consistent age stacking based on the youngest single grain approach would appear to support the interpretation of *Fildani et al.* [2009], the ages with stated 2σ errors for the TuffZirc population are also compatible with the known stratigraphic sequence. This is shown by Monte Carlo simulations, which provide a "best fit" age model for stratigraphically ordered samples (Table 3). For the purposes of this paper, the 266–276 Ma age range identified by the Monte Carlo "best fit" ages is regarded as the depositional age interval of the Ecca Group (Figure 11). This age range agrees with the biostratigraphic data for vertebrate fauna, i.e., the *Eodicynodont* zonation of the Ecca Group-Beaufort Group contact [*Rubidge et al.*, 1999], as well as new palynologic data for the Ecca Group [*Barbolini*, 2014].

The hypothesis of undetected Pb-loss in the zircons that yielded "young" ages underpins this reinterpretation of the published age data set, and requires further explanation and substantiation. Notably, all of the problematic data were obtained almost exclusively from ashbeds in the most deeply buried sections of the middle Ecca Group, where illite crystallinity data reported by *Hälbich et al.* [1983] suggest low-grade metamorphic temperatures in the range of 170–200°C. This is the temperature range for which fluid-mediated lattice damage begins to occur in zircon [*Geisler-Wierwille et al.*, 2002; *Geisler et al.*, 2007], so our contention of undetected Pb-loss has a firm experimental basis. We followed two lines of inquiry to substantiate the Pb-loss hypothesis. First, we searched for evidence of Pb-loss by looking for systematic differences between the $^{235}U/^{207}Pb$ age (t_{207}) and the $^{238}U/^{206}Pb$ age (t_{206}) in zircon grains. Discordance caused by Pb loss affects the two U/Pb isotopic systems differently because of different relative abundances of parent-daughter isotopes, with discordance typically marked by a trend of $t_{207} > t_{206}$. No such trend is evident from the "young" population, mostly due to the large errors associated with t_{207} based on analytical uncertainty in the measured abundance of ^{207}Pb. The second approach seeks to differentiate the "too-young" population on the basis of chemistry, specifically, the levels of U and Th. Because Pb-loss in zircon is generally considered to result from damage to the crystal lattice through accumulated radiation dosage, we would expect metamictization in the youngest grains to result from higher U and Th content due to higher accumulated radioactive dosages. To test this, we examined the U and Th contents of the youngest grains, which were defined as those that lie outside the youngest TuffZirc population mode. For comparison, the U and Th contents of the concordant, <300 Ma grains were also examined, as well as the discordant zircon populations that were reported but not interpreted for age information by *Fildani et al.* [2007, 2009]. Not surprisingly, the highest combined U and Th concentrations are found in the discordant population (n = 52), with a median value of 1200 +380/−360 ppm (95% confidence). The zircon population that is equivalent in age to the TuffZirc age or older has a median combined U and Th content of 790 +53/−120 ppm (95% conf.), significantly lower than the discordant data set. The "young" zircon population has a median U+Th content of 868 +110/−130 (95% conf.), and the youngest sample (CVX12) reported by *Fildani et al.* [2009] has a combined U+Th content of 1167 +470/−250 ppm (95% conf.). Taken together, these observations suggest a systematic shift toward more radioactive content for the anomalously young population of zircon grains (Figure 11), reinforcing our suspicion that undetected Pb-loss caused by increasing radiation dosage and attendant metamictization has affected this population. Confirmation of this Pb-loss

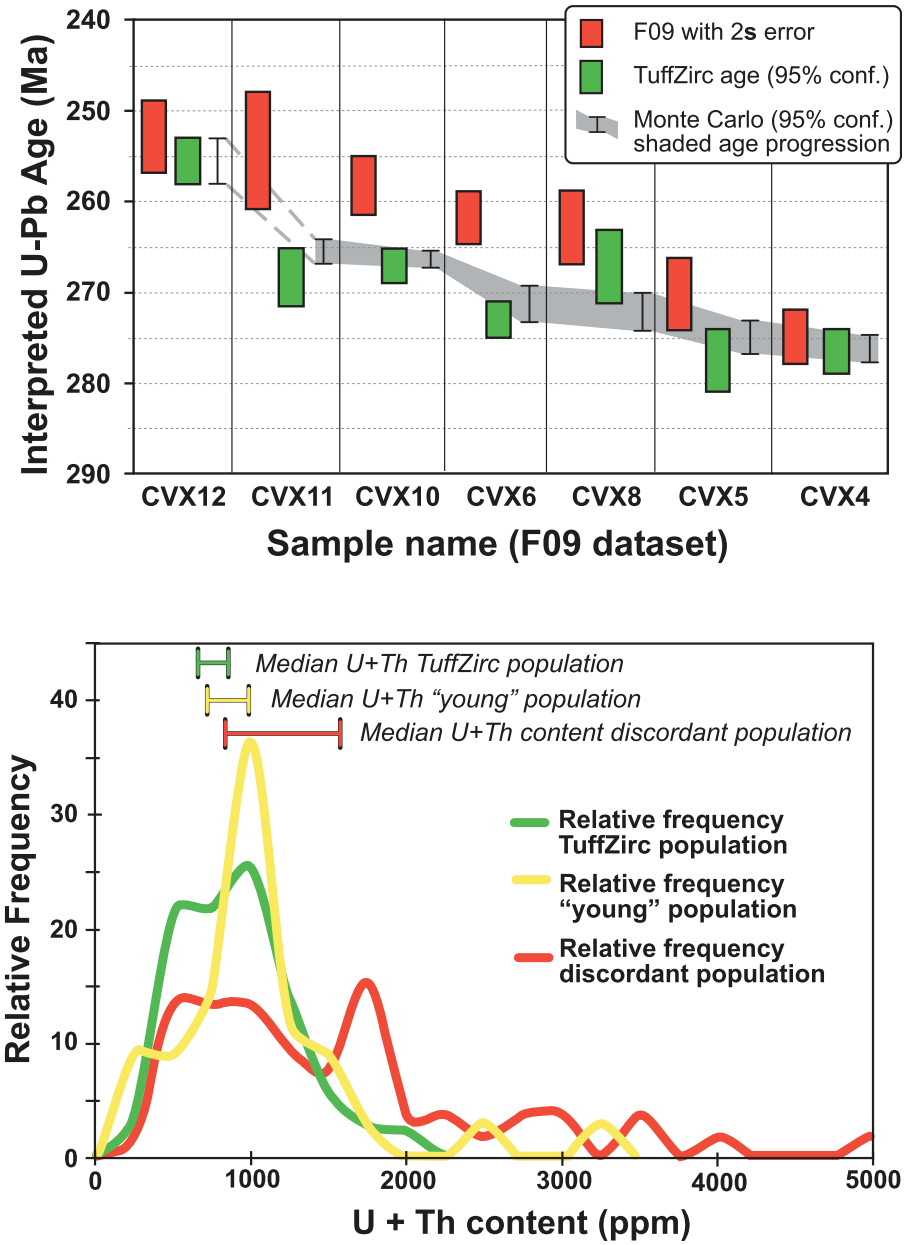


Figure 11. (top) Calculated TuffZirc ages (green) from Ecca Group U-Pb data reported by *Fildani et al.* [2009] demonstrate consistently older ages identified from coherent populations of zircon versus original interpretation (red, F09). Samples are ordered stratigraphically with the youngest to the left. Gray swath indicates the best fit ages of 265–275 Ma determined by Monte Carlo simulation. (bottom) Histogram of U+Th content for different populations; TuffZirc selected populations (green), "young" grains used as the basis of the *Fildani et al.* [2009] age interpretation (yellow), and discordant analyses that were reported but not age-interpreted by *Fildani et al.* [2009]. Note the shift toward more radioactive content for the "young" population, consistent with the undetected Pb-loss that we postulate.

hypothesis requires thermal ionization mass spectrometry analysis of these grains, preferably preceded by chemical abrasion.

6. Conclusions

The paleomagnetic directions and rock-magnetism of the upper part of the Abrahamskraal Formation in the lower Beaufort Group studied in the composite, 600 m thick Great Wall, Link, and Bloukrans sections are virtually identical to that studied at the Ouberg Pass. Intermediate and high-temperature components (A and B, respectively) of the NRM have been isolated at both locations at the same temperature ranges and

their directions are perfectly overlapping. The intermediate temperature component-A is interpreted to result from regional heating caused by the emplacement of widespread Karoo intrusions in the Early Jurassic. Paleomagnetic directions are consistent with previous work and increase the available data contributing to a Late-Permian geomagnetic pole position for the African plate. The magnetostratigraphy of this composite section indicates the prevalence of normal polarity magnetozones, which is clearly different from the dominantly reversed polarity of the Ouberg Pass site. We present two alternative interpretations: hypothesis 1 assigns a Capitanian age (ca. 266–260 Ma) to the GK-LK-BK composite section, providing a correlation with the Ouberg Pass stratigraphy. This interpretation requires an order of magnitude increase in sedimentation rate for GW-LK-BK section. Hypothesis 2 posits a correlation to the normal polarity zones that predominate during the middle Wuchianpingian to middle Changshingian times (ca. 256–254 Ma). Both interpretations contradict are inconsistent with recent interpretations of a late Permian age for the underlying Ecca Group. Our re-examination of the U/Pb zircon data from ashbeds used for this late Permian age interpretation for the upper Ecca Group indicates higher U+Th content in putatively latest Permian grains, highlighting the potential for Pb loss from radiation-damaged grains. Monte Carlo modeling of TuffZirc populations indicates an age range of 266–276 Ma for deposition of the upper Ecca Group.

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